

Nested-quotient temporal histories and homological flat bands in strong-coupling $SU(N)$ Hamiltonian lattice gauge theory

Master derivation, evidence, and scope document

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Abstract

We study the Kogut–Susskind Hamiltonian for pure $SU(N)$ gauge theory on a periodic cubic spatial lattice in the strong-coupling regime. After selecting the complete, gauge-invariant, charge-odd one-plaquette eigenspace, fixed-order contributions are organized as ordered magnetic histories on the Haar–Gram quotient of reachable Wilson networks. We prove self-containedly that the complete momentum dependence through second order is the oriented edge–face Gram operator $t_N u^2 B(\mathbf{k}) B(\mathbf{k})^\dagger$, where $B = \partial_2^\dagger$ and

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0, \quad N \geq 3.$$

Up to a common scalar, the projected spectrum is $\{0, t_N u^2 q, t_N u^2 q\}$, with $q = 4 \sum_j \sin^2(k_j/2)$. The flat carrier is $\ker \partial_2$: elementary cube boundaries give compact localized states, while three wrapping sheets complete the carrier on the periodic L^3 complex, giving exact dimension $L^3 + 2$ for $L \geq 3$.

For $SU(3)$, supplied exact-arithmetic derivation ledgers report

$$H_{\text{eff},-} = E_{\text{flat}}(u)I + \left(\frac{5}{612}u^2 + \frac{1975}{124848}u^3 \right) BB^\dagger + O(u^4),$$

where $E_{\text{flat}} = 8/3 + u + 11u^2/306 - 109151u^3/249696$, together with a zero census for 32 determinant-aware lifter classes. The present bundle does not contain the raw class payload or a complete independent reconstruction of that linked ledger. Conditional on its exhaustiveness and correctness, the carrier remains dispersionless through third order.

A supplied symbolic ledger further reports the cube formula $\alpha_N = 640/[N(N^2 - 1)^3]$, and a supplied microscopic ledger reports its equality with the complete axial coefficient for $N = 3, \dots, 12$. The bundled verifier checks arithmetic consequences of the encoded formula but does not reconstruct either underlying enumeration. Conditional on those records, $\alpha_3 = 5/12$ gives nonzero

axial dispersion at fourth order. One mixed off-axis coefficient remains open, so no complete fourth-order band or bandwidth is claimed. All statements concern a formal finite-order projected sector, not a continuum glueball or mass-gap theorem.

1 Introduction

Hamiltonian strong-coupling expansions organize lattice gauge dynamics around the electric vacuum and its flux excitations [1, 2, 3]. In three spatial dimensions, a fundamental plaquette excitation carries one of three face orientations. At second order, neighboring faces communicate through a shared link, so the projected problem resembles a three-orbital tight-binding model. The signs of those matrix elements, however, are fixed by the orientation of the plaquette boundary and by charge conjugation. The resulting interference is the central object of this paper.

Flat bands generated by incidence or Gram matrices have a long history. Local topology, line-graph constructions, compact localized states, and the need for noncontractible states on periodic systems are well established [4, 5, 6, 7, 8]. Rectangular factorizations and index arguments provide a general language for such zero modes [9]. Closely related curl-free or closed-surface structures occur in two-form spin liquids [10], and a recent independent construction treats face-graph and higher-cell flat bands directly [12]. We therefore do not claim that incidence flatness, cellular compact localized states, or homological counting are new in isolation.

The result here is instead dynamical. Starting from non-Abelian $SU(N)$ Hamiltonian lattice gauge theory, exact strong-coupling matrix elements generate an *oriented edge–face* Gram operator in the charge-odd one-plaquette sector. Group representation theory fixes its coefficient, while the cubical chain complex fixes its kernel and spectrum. The separation is exact:

$$\text{color dynamics} \longrightarrow t_N, \quad \text{cellular geometry} \longrightarrow BB^\dagger.$$

This produces three self-contained results and two ledger-dependent conclusions.

1. For every $N \geq 3$, the complete momentum dependence at order u^2 is $t_N u^2 B B^\dagger$, with the positive rational function t_N displayed in the abstract.
2. The Bloch spectrum of $B B^\dagger$ is $\{0, q(\mathbf{k}), q(\mathbf{k})\}$. The zero branch is the space of boundary-free face amplitudes.
3. On the periodic L^3 cubical complex, the zero-mode count is $L^3 + 2$: $L^3 - 1$ independent cube boundaries and three harmonic wrapping sheets.
4. For $SU(3)$, supplied exact-arithmetic ledgers report that all 32 candidate third-order lifter classes vanish and report the scalar and shared-link coefficients displayed in Section 5. If that census is exhaustive and the linked ledger is complete, the effective Hamiltonian remains in the incidence algebra through order u^3 .
5. Supplied ledgers report the all-rank cube formula $\alpha_N = 640/[N(N^2 - 1)^3]$ and its equality with the complete axial coefficient for $N = 3, \dots, 12$. Conditional on those calculations, the $SU(3)$ carrier acquires nonzero axial dispersion at order u^4 . The complete off-axis operator remains open.

Items 1–3 are derived within this manuscript. Items 4–5 are conditional computational conclusions: the exact rationals are recorded in the supplied derivation ledgers, but the raw class contractions, symbolic history certificate, and microscopic rank-sweep artifacts are not included in the present bundle. Repeated assertions in synthesis documents are not treated as independent reproductions.

These statements concern the effective Hamiltonian projected to the degenerate one-plaquette manifold. Virtual multi-plaquette states are retained in the resolvents, but the external spectral problem is not the full gauge-theory Hilbert space. At $\mathbf{k} = 0$ the three incidence branches touch; the finite-volume separation collapses as L^{-2} . Nothing below proves convergence of the strong-coupling series, persistence under unrestricted higher-order operators, identification with the lightest physical glueball, or a continuum mass gap.

2 Hamiltonian and projected sector

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$, $L \geq 3$, be a cubic spatial lattice with periodic boundary conditions. In units of the electric coupling, the Kogut–Susskind Hamiltonian is

$$H_\beta = \frac{1}{2} \sum_\ell C_2(\ell) + \beta_N \sum_p \left(1 - \frac{1}{N} \operatorname{Re} \operatorname{Tr} U_p\right). \quad (1)$$

After dropping the additive constant, write

$$H = H_E - uV, \quad V = \sum_p (\chi_p + \bar{\chi}_p), \quad u = \frac{\beta_N}{2N}, \quad (2)$$

where $\chi_p = \operatorname{Tr} U_p$ is the fundamental character. For $SU(3)$, $u = \beta_3/6 = 1/g_H^4$ in the convention used below. The definition $u = \beta_N/(2N)$, rather than the $SU(3)$ -specific final equality, is the all-rank coupling convention.

All quoted excitation energies are formed from the linked, vacuum-subtracted operator

$$H_{\text{exc}}(u) = H(u) - E_{\text{vac}}(u)I. \quad (3)$$

This subtraction changes only the scalar ledger; incidence hopping and carrier shape are unaffected.

The electric vacuum $|0\rangle$ has the trivial representation on every link. For $N \geq 3$, normalized one-plaquette states of definite charge conjugation may be chosen as

$$|p, \pm\rangle = \frac{\chi_p \pm \bar{\chi}_p}{\sqrt{2}} |0\rangle, \quad (4)$$

up to an immaterial phase in the odd combination. Character orthogonality gives unit norm. Each of the four boundary links carries the fundamental or antifundamental representation, so the unperturbed energy is

$$E_{F,N} = 4 \left(\frac{1}{2} C_F \right) = 2C_F = \frac{N^2 - 1}{N}, \quad C_F = \frac{N^2 - 1}{2N}. \quad (5)$$

For $SU(2)$, complex conjugation is an inner gauge transformation and the charge-odd state vanishes; this is why the construction begins at $N = 3$.

We work from this point inside the charge-odd fundamental Wilson-loop cyclic sector with trivial electric one-form center flux around each torus cycle (the contractible-loop sector). Let $\mathbf{1}_-$ be its identity and let P_- be the full $E_{F,N}$ spectral projector in this sector. For $L \geq 3$, its range is the charge-odd one-plaquette span: a reduced contractible fundamental loop at the displayed electric energy has four distinct links and is an elementary plaquette. This sector qualification also excludes the accidentally degenerate straight wrapping loop when $L = 4$. Set $Q = \mathbf{1}_- - P_-$. Thus the reduced resolvent crosses no omitted $E_{F,N}$ eigendirection; any additional degenerate direction in a different regulator would have to be added to P_- . Degenerate perturbation theory for Equation (3) gives a formal effective operator

$$H_{\text{eff},-} = P_- H_E P_- + uH_1 + u^2 H_2 + u^3 H_3 + \dots, \quad (6)$$

where every H_r is linked and vacuum-subtracted and contains the appropriate Q -space resolvents. Thus projection fixes the external states but does not discard virtual multi-plaquette sectors. Translation invariance reduces the external problem at each $\mathbf{k}$ to three face orbitals, which we order as (12), (13), (23).

2.1 Ordered physical histories

The static set of plaquettes used by a process is not its complete dynamical label. Write $V = \sum_f W_f$, so the physical perturbation is $-u \sum_f W_f$. An order- r grouped face history is an ordered word

$$h = (f_1, \dots, f_r). \quad (7)$$

It carries a matrix element of the schematic form, with $W_j = W_{f_j}$,

$$\mathcal{A}[h] = \langle q_{\text{out}} | (-W_r) R \cdots R(-W_1) | q_{\text{in}} \rangle. \quad (8)$$

Each $W_f = \chi_f + \bar{\chi}_f$ sums the two orientations; when the amplitude is expanded, the orientation choice is retained as part of the intermediate physical-state label. Thus an expanded word is $\tilde{h} = ((f_1, \sigma_1), \dots, (f_r, \sigma_r))$, with static signed support $\pi(\tilde{h}) = \sum_j \sigma_j e_{f_j}$. In general,

$$\pi(\tilde{h}) = \pi(\tilde{h}') \not\Rightarrow \mathcal{A}[\tilde{h}] = \mathcal{A}[\tilde{h}']. \quad (9)$$

Here

$$R = Q(E_{F,N} - H_E)^{-1} Q \quad (10)$$

is nonsingular on the stated fixed-order physical complement. Proper returns to the retained space are instead represented by the corresponding energy-dependent Feshbach folds. Two orderings with the same final signed face support can traverse different intermediate representations and therefore acquire different electric denominators. Temporal ordering cannot in general be replaced by a static cellular circuit.

On a bounded support S , let $\mathcal{W}_{S,r}$ be the span of Wilson networks reachable through order r , and let G_S be its exact Haar Gram matrix. Resolvents act on the physical quotient

$$\mathcal{H}_{S,r}^{\text{phys}} = \mathcal{W}_{S,r} / \ker G_S, \quad (11)$$

so Gram-null trace identities are removed before electric denominators are applied. State labels must retain any representation multiplicity or intertwiner information needed to specify a physical vector; an energy label alone need not be sufficient.

Proposition 1 (Fixed-order local support). *Give local face operators the adjacency relation induced by overlapping on an elementary link, and let R_0 be the finite root support. On a locally finite support graph, every connected rooted contribution with at most r insertions lies in the radius- r neighborhood $\text{Star}_r(R_0)$. If every bounded support has a finite dimensional reachable quotient Equation (11), then, after choosing a basis in each quotient, the candidate quotient-basis labels and fixed-order Feshbach monomials form a finite universe.*

Proof. After linked-cluster reduction, the union of the root and inserted supports is connected. Reaching a face at graph distance d requires at least d inserted supports along a connecting path, so $d \leq r$. Local finiteness makes this neighborhood finite. The assumed finite dimensional reachable quotients then admit finite bases, while a formal Feshbach expansion through fixed order contains only finitely many powers of the perturbation, energy derivative, and reduced resolvent. $\square$

Partial histories that arrive at the same physical quotient vector with the same rooted support and fold metadata can be merged by linearity. The proposition is a

completeness statement at each fixed order, not a closed formula for the first dispersive order and not a guarantee that an order-by-order search terminates if a band is flat to all orders.

2.2 Finite folds and rooted scalar assembly

Let $\delta E = E - E_{F,N}$ and $\mathcal{U}(u) = Q(H - H_E)Q = -uQVQ$. The exact Feshbach map uses

$$\mathcal{F}(E, u) = P_- H P_- + P_- H Q (E - Q H Q)^{-1} Q H P_-.$$

At any specified total order r , its resolvent has the formal expansion

$$(E - Q H Q)^{-1} = R \sum_{n=0}^r [(\mathcal{U} - \delta E) R]^n + O_{\text{tot}}(r+1). \quad (12)$$

Because δE has positive perturbative valuation, only finitely many resolvent powers and energy-derivative fold labels occur at fixed order. A propagation key may therefore be written as (a, S, ϕ) , where a is a chosen quotient-basis label; accumulated partial amplitudes form a finite coordinate vector indexed by a , $S \subseteq \text{Star}_r(R_0)$ is rooted support, and ϕ records the finite fold skeleton. Under the hypotheses of Proposition 1, exact input matrices reduce every specified coefficient to finite algebra on these states. This is a fixed-order algebraic reduction, not an all-order proof of flatness.

Scalar assembly is a separate rooted linked-cluster operation. If $\delta_r(C)$ is the inclusive coefficient on a labeled connected rooted support C , after support-union convolution, retained-space folds, and vacuum subtraction, define

$$\omega_r(C) = \delta_r(C) - \sum_{\substack{S \subseteq C \\ S \text{ rooted, connected}}} \omega_r(S). \quad (13)$$

When supports are grouped by isomorphism class, the corresponding embedding multiplicities must be inserted in the sum. This support-poset inversion fixes the scalar/rest coordinate; it is logically independent of the carrier-shape compression used below.

3 Oriented incidence and the exact carrier

The periodic cubical complex has the cellular chain sequence

$$C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0, \quad \partial_2 \partial_3 = 0. \quad (14)$$

The one-plaquette amplitudes are elements of C_2 . Their signed amplitude leaking into links is $\partial_2 c$. This is a cellular closed-surface condition, not the ordinary microscopic vertex Gauss law: Wilson-loop trace contractions already

enforce microscopic gauge invariance. The chain identity $\partial_1 \partial_2 = 0$ is only its cellular analogue.

For

$$d_j = e^{ik_j} - 1, \quad q(\mathbf{k}) = \sum_{j=1}^3 |d_j|^2 = 4 \sum_{j=1}^3 \sin^2 \frac{k_j}{2}, \quad (15)$$

choose the adjoint, link-to-face Bloch incidence matrix

$$B(\mathbf{k}) = \partial_2(\mathbf{k})^\dagger = \begin{pmatrix} d_2 & -d_1 & 0 \\ d_3 & 0 & -d_1 \\ 0 & d_3 & -d_2 \end{pmatrix}. \quad (16)$$

Orbital gauge changes multiply rows or columns by phases and do not alter the following statements. The physical face-to-link leakage map is $B^\dagger = \partial_2$, so the carrier condition is $B^\dagger c = 0$.

Theorem 2 (Incidence factorization). *Let*

$$\mathbf{w}(\mathbf{k}) = (\bar{d}_3 \quad -\bar{d}_2 \quad \bar{d}_1)^T. \quad (17)$$

Then

$$BB^\dagger = qI - \mathbf{w}\mathbf{w}^\dagger, \quad B^\dagger \mathbf{w} = 0, \quad (18)$$

and therefore

$$\text{spec}(BB^\dagger) = \{0, q, q\}. \quad (19)$$

If $S = BB^\dagger - 4I$ is the signed shared-link adjacency, then

$$\text{spec } S(\mathbf{k}) = \{-4, q(\mathbf{k}) - 4, q(\mathbf{k}) - 4\}. \quad (20)$$

Proof. Multiplying Equation (16) by its adjoint gives the first identity in Equation (18) entry by entry. Since $\|\mathbf{w}\|^2 = q$, the rank-one operator $qI - \mathbf{w}\mathbf{w}^\dagger$ annihilates $\mathbf{w}$ and acts as qI on its orthogonal complement. Subtracting $4I$ gives Equation (20). $\square$

For $\mathbf{k} \neq 0$, the carrier is one dimensional and is represented by $\mathbf{w}(\mathbf{k})$. At Γ , $B = 0$, so all three branches meet. The normalized Bloch eigenvector has no direction-independent limit at Γ ; this is the familiar singular-flat-band mechanism behind the incompleteness of a purely compact localized basis [6, 7].

Theorem 3 (Finite-torus carrier). *For the periodic cubic cellulation of T^3 ,*

$$\ker \partial_2 = \text{im } \partial_3 \oplus \mathcal{H}_2, \quad (21)$$

with

$$\dim \text{im } \partial_3 = L^3 - 1, \quad \dim \mathcal{H}_2 = 3. \quad (22)$$

Hence

$$\boxed{\dim \ker \partial_2 = L^3 + 2.} \quad (23)$$

Elementary cube boundaries span $\text{im } \partial_3$; three independent wrapping sheets span $\mathcal{H}_2 \simeq H_2(T^3)$.

Proof. There are L^3 three-cells and no four-cells. Since $b_3(T^3) = 1$, rank-nullity gives $\text{rank } \partial_3 = L^3 - 1$. Discrete Hodge decomposition [11] gives $Z_2 = \ker \partial_2 = B_2 \oplus \mathcal{H}_2$, and $\dim \mathcal{H}_2 = b_2(T^3) = 3$. Adding the dimensions proves Equation (23). The single relation among translated elementary cube boundaries is the oriented sum over all cubes. $\square$

The number $L^3 + 2$ is the nullity of the down boundary Laplacian, not a Betti number. The Betti contribution is only the final 3. For the ordinary twelve-neighbor Bloch convention we assume $L \geq 3$; at $L = 2$, coincident periodic neighbors require an explicit multigraph convention. The first nonzero incidence eigenvalue is

$$q_{\min} = 4 \sin^2 \frac{\pi}{L}, \quad (24)$$

which will control the coefficient-level finite-volume separation.

4 Exact all-rank dynamics at second order

We now derive the coefficient multiplying the geometrical operator. Two plaquettes sharing a link leave six non-shared fundamental links after the external contractions. On the shared link, the possible fusion channels depend on the relative induced orientation:

$$F \otimes \bar{F} = \mathbf{1} \oplus \text{Adj}, \quad F \otimes F = \Lambda^2 F \oplus \text{Sym}^2 F. \quad (25)$$

Let ρ be one of these irreducible channels. Isotropy of the normalized shared-link tensor assigns squared norm d_ρ/N^2 to the channel projector. The six nonshared half-links contribute $3C_F$ and the fused link contributes $C_\rho/2$, whereas the external one-plaquette energy is $2C_F$. The resolvent weight is therefore

$$w_\rho = -\frac{d_\rho/N^2}{C_F + C_\rho/2}. \quad (26)$$

The required dimensions and Casimirs are

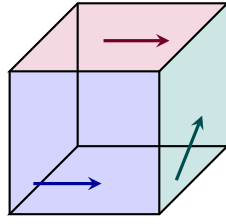
$$\begin{aligned} d_{\mathbf{1}} &= 1, & C_{\mathbf{1}} &= 0, & d_{\text{Adj}} &= N^2 - 1, & C_{\text{Adj}} &= N, \\ d_{\Lambda^2 F} &= \frac{N(N-1)}{2}, & C_{\Lambda^2 F} &= \frac{(N+1)(N-2)}{N}, \\ d_{\text{Sym}^2 F} &= \frac{N(N+1)}{2}, & C_{\text{Sym}^2 F} &= \frac{(N-1)(N+2)}{N}. \end{aligned} \quad (27)$$

Substitution into Equation (26) and summation within the two orientation families gives

$$\mathcal{A}_N = -\frac{2N^3}{(N^2 - 1)(2N^2 - 1)}, \quad (28)$$

$$\mathcal{B}_N = -\frac{4N(N^2 - 2)}{(N^2 - 1)(4N^2 - 9)}. \quad (29)$$

The charge-odd combination takes their signed difference.



cube boundary $\partial_3 c$
six oriented faces

$$C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \rightarrow \ker \partial_2 = \text{im } \partial_3 \oplus \mathcal{H}_2$$

$$\partial_2 \partial_3 = 0 \quad (L^3 - 1) \text{ cube boundaries} \\ \text{local carrier} \quad + 3 \text{ wrapping sheets}$$

Figure 1: The cellular origin of the carrier. An oriented elementary cube boundary is a compact localized state because every edge occurs twice with opposite induced signs. On a torus, cube boundaries span $\text{im } \partial_3$, but three harmonic wrapping sheets are also required.

Theorem 4 (All-rank shared-link law). *For every integer $N \geq 3$, the charge-odd one-plaquette effective Hamiltonian has the form*

$$H_{\text{eff},-}^{(N)}(\mathbf{k}, u) = E_{\text{flat},N}(u)I + t_N u^2 B(\mathbf{k})B(\mathbf{k})^\dagger + O(u^3), \quad (30)$$

where the scalar $E_{\text{flat},N}$ is momentum independent and

$$t_N = \mathcal{B}_N - \mathcal{A}_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0. \quad (31)$$

The three bands through second order are

$$E_{\text{flat},N}(u), \quad E_{\text{flat},N}(u) + t_N u^2 q(\mathbf{k}) \quad (\text{twice}). \quad (32)$$

Proof. The diagonal and within-plaquette terms are scalars in face-orbital space. Every second-order inter-plaquette process uses one shared link, and its relative orientation is precisely the incidence product appearing in $S = BB^\dagger - 4I$. The contributions in Equations (28) and (29) therefore assemble as $d_N I + t_N S$. Absorbing $-4t_N$ into the scalar gives Equation (30). The factorization theorem then gives Equation (32). For $N \geq 3$, every denominator in Equation (31) is positive and $N^2 - 4 > 0$, proving strict positivity. $\square$

Two useful consequences are

$$t_3 = \frac{5}{612}, \quad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} - \frac{77}{64N^7} + O(N^{-9}). \quad (33)$$

Let $q_{\text{max}}(L)$ denote the largest sampled value of $q(\mathbf{k})$ on the periodic L^3 momentum grid. Then

$$q_{\text{max}}(L) = \begin{cases} 12, & L \text{ even,} \\ 12 \cos^2(\pi/(2L)), & L \text{ odd.} \end{cases} \quad (34)$$

Consequently the finite-volume width of the *full retained charge-odd manifold* is

$$W_{N,L}^{(-)} = q_{\text{max}}(L)t_N u^2 + O(u^3). \quad (35)$$

For even L , and in the continuous-zone or thermodynamic limit, the quadratic coefficient obeys $[u^2]W_{N,L}^{(-)} = 12t_N \sim$

$3/N^3$ as $N \rightarrow \infty$. This N^{-3} statement concerns the stiffness of the dispersive doublet and the full-manifold width. It is not motion of the lowest branch, which remains flat at this order.

For comparison, removing the orientation signs gives an unsigned incidence matrix $N(\mathbf{k})$ with $A + 4I = NN^\dagger$. Its adjacency spectra at Γ and R are $\{12, 0, 0\}$ and $\{-4, -4, -4\}$, respectively. No momentum-independent eigenvalue can belong to both sets. The flat branch is therefore not a generic consequence of three face orbitals or cubic symmetry; the charge-odd orientation signs are essential.

5 Reported $SU(3)$ third-order ledger

We now specialize to $SU(3)$. The rational coefficients in this section are transcribed from supplied exact-arithmetic derivation ledgers. `verify_core.py` checks relations among those rationals, and the supplied local resolvent program independently reconstructs the limited kernels stated below. Neither check reconstructs the complete linked third-order history universe. We therefore separate reported inputs from their conditional spectral consequence. The supplied within-plaquette ledger reports

$$\frac{8}{3} + u + \frac{1}{2}u^2 + \frac{7}{32}u^3. \quad (36)$$

At second order, shared-link self-energy and hopping terms reduce the flat coefficient to

$$\frac{1}{2} + 12 \left(-\frac{11}{306} \right) - 4 \left(\frac{5}{612} \right) = \frac{11}{306}, \quad (37)$$

when organized in the scalar-plus- $BB^\dagger$ convention. Equivalently, the charge-odd diagonal and signed-hop coefficients before the shift are $7/102$ and $5/612$.

At third order, processes confined to one plaquette or to a two-plaquette domino renormalize the same scalar and shared-link structures. A new connected geometry could involve three distinct plaquettes.

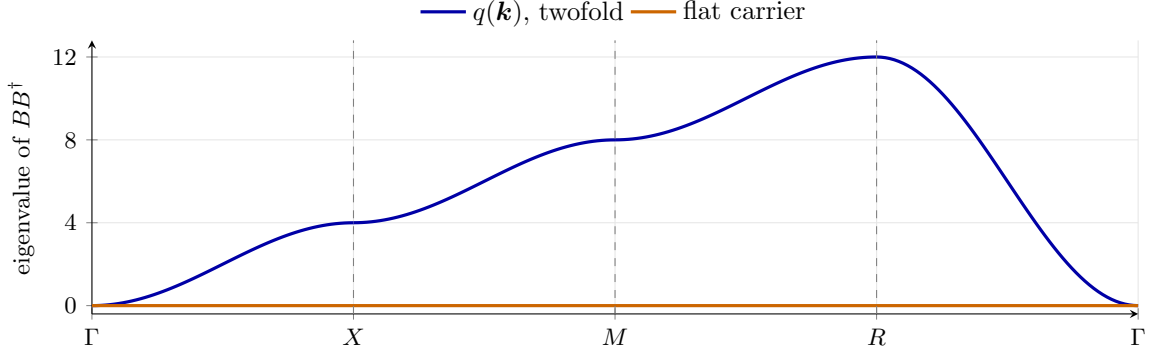


Figure 2: Universal incidence spectrum along $\Gamma-X-M-R-\Gamma$. Color dynamics only rescales the blue doublet by $t_N u^2$, or by $\tau(u)$ for the ledger-dependent $SU(3)$ third-order result. All three branches touch at Γ .

Reported computational result 5 (Third-order lifter census). *The supplied derivation ledger reports that the connected three-distinct-plaquette lifters form 32 determinant-aware five-trace classes and that the charge-odd $SU(3)$ contraction of every class is zero. The present bundle does not contain the raw class definitions, per-class contraction outputs, or an executable enumeration establishing exhaustiveness. This statement records the ledger result; it is not independently proved by the portable verifier.*

Remark 6. The reported census is deliberately scoped to the new three-distinct-plaquette lifters. Repeated-plaquette and domino processes do not vanish; they produce the nonzero scalar and hopping coefficients below. The supplied ledger states that its enumeration groups the candidate lifters into 32 five-trace numerator classes and returns zero for every class. A bare “unmatched link” mnemonic is not a proof at $SU(3)$, because three equally oriented fundamental factors can couple through an epsilon tensor; the reported contraction includes these finite-rank determinant sectors.

The same ledger reports the nonzero third-order per-neighbor quantities

$$b_3 = \frac{1975}{124848}, \quad \text{leak}_3 = -\frac{12331}{249696}. \quad (38)$$

Combining them with the tower coefficient yields

$$d_3 = \frac{7}{32} + 12 \text{leak}_3 - 4b_3 = -\frac{109151}{249696}. \quad (39)$$

An independent exact local resolvent reconstruction supplies a useful partial check. It obtains

$$\begin{aligned} \text{spec } H_E(F \otimes F) &= \left\{ \frac{14}{3}, \frac{17}{3} \right\}, \\ \text{spec } H_E(F \otimes \bar{F}) &= \left\{ 4, \frac{11}{2} \right\}, \end{aligned} \quad (40)$$

and, writing $W = V$, $P = P_-$, while the physical pertur-

bation is $-uW$,

$$\begin{aligned} PWRWP|_S &= \frac{5}{612}S, \\ PWR^2WP|_S &= -\frac{1975}{124848}S. \end{aligned} \quad (41)$$

Here $|_S$ denotes the coefficient of the signed-adjacency operator S , not a restriction to a new state subspace. Separately, the exact $SU(3)$ character identity gives $PWP = -P$. The sign in the physical third-order fold is reversed by the three factors of $-W$, giving $+1975S/124848$. Since the recorded physical total has the same coefficient, subtraction would give a zero direct S component if both are expressed in the same effective-Hamiltonian gauge. This is a conditional consistency consequence, not a cold direct evaluation. The reconstruction imports the complete third-order coefficient only as a regression target and does not cold-evaluate the direct $PWRWRWP$ term. Thus Equation (41) independently reconstructs the second-order kernel and squared-resolvent fold, but it is not a second derivation of the reported total and does not establish that the 32-class census is exhaustive.

Conditional corollary 7 ($SU(3)$ third-order factorization). *Assume that the census reported in Reported computational result 5 exhausts every connected third-order contribution capable of leaving the scalar-plus-shared-link incidence algebra, that its determinant-aware contractions are correct, and that the reported linked coefficients in Equation (38) are complete and expressed in the same effective-Hamiltonian convention. Then, in the charge-odd one-plaquette sector of $SU(3)$,*

$$H_{\text{eff},-}(\mathbf{k}, u) = E_{\text{flat}}(u)I + \tau(u)B(\mathbf{k})B(\mathbf{k})^\dagger + O(u^4), \quad (42)$$

where

$$E_{\text{flat}}(u) = \frac{8}{3} + u + \frac{11}{306}u^2 - \frac{109151}{249696}u^3, \quad (43)$$

$$\tau(u) = \frac{5}{612}u^2 + \frac{1975}{124848}u^3. \quad (44)$$

and the three formal bands through $O(u^3)$ are

$$E_{\text{flat}}(u), \quad E_{\text{flat}}(u) + \tau(u)q(\mathbf{k}) \quad (\text{twice}). \quad (45)$$

Proof. The second-order result is Theorem 4 at $N = 3$. Under the stated census assumptions, no new third-order Bloch shape survives. The reported remaining scalar leakage and signed shared-link hop assemble by Equation (39); that identity is an exact arithmetic check. The self-contained incidence factorization then gives the conditional spectrum. $\square$

Under the assumptions of Conditional corollary 7, the finite-volume coefficient-level gap between the carrier and the nearest dispersive state is

$$\Delta_L(u) = 4\tau(u) \sin^2 \frac{\pi}{L} + O(u^4). \quad (46)$$

For sufficiently small positive u , the displayed coefficient is positive. Since it scales as L^{-2} , it does not provide volume-uniform spectral isolation.

Hamer's independent 1989 Hamiltonian series for the 1^{+-} channel [3] supplies a non-load-bearing cross-check of the rest-frame coefficients. With his $x = 2u$ convention and the Hamiltonian normalization $H_{\text{here}} = W_{\text{Hamer}}/2$, the converted coefficients through x^3 agree with Equation (43) to the precision printed in that paper; the cubic comparison differs by approximately 1.1×10^{-12} . The analytic assignment of the Γ -point orbital triplet to T_1^{+-} is consistent with that comparison. We nevertheless reserve the term *glueball mass* for a full-sector spectral statement, which is not proved here.

6 What the homology protects

Three notions must be kept separate. A momentum-independent scalar re-anchoring changes the rest coordinate, a non-scalar retained operator changes internal band shape, and only momentum dependence establishes dispersion. On the finite-torus carrier $Z = \ker \partial_2$, with orthogonal projector P_Z , scalar-free action is measured by

$$\mathfrak{M}_Z(A) = P_Z A P_Z - \frac{\text{tr}_Z(P_Z A P_Z)}{\dim Z} P_Z. \quad (47)$$

This map annihilates both constant scalars and every $\partial_2^\dagger X \partial_2$ term. It diagnoses non-scalar action on the global carrier, but is not by itself a transport criterion: a momentum-independent onsite multiplet splitting can be non-scalar, while a one-dimensional Bloch fiber is automatically scalar. In the setting used below, every lower-order carrier correction preceding a candidate order is a momentum-independent scalar, so the applicable coefficientwise criterion is

$$r_{\text{disp}} = \min \left\{ r : \text{spec}(P_{\text{car}}(\mathbf{k}) H_r(\mathbf{k}) P_{\text{car}}(\mathbf{k})) \text{ depends on } \mathbf{k} \right\}. \quad (48)$$

This isolated-coefficient test is valid here because of the stated lower-order scalar premise. In a general multiband

problem with an earlier non-scalar splitting, r_{disp} must instead be read from the formal eigenvalue multiset of $\sum_{j \leq r} u^j P_{\text{car}} H_j P_{\text{car}} \bmod u^{r+1}$, using recursive degenerate perturbation theory; the spectrum of an isolated H_r is then not invariant. Momenta at which the carrier dimension jumps, notably Γ , are handled by directional limits or by the full three-orbital spectrum.

The lower-order result extends immediately to any correction that remains inside the boundary-factorized operator class.

Proposition 8 (Boundary-factorized rigidity). *Suppose an order- r Bloch correction has the form*

$$H_r(\mathbf{k}) = a_r I + b_r S(\mathbf{k}) + B(\mathbf{k}) M_r(\mathbf{k}) B(\mathbf{k})^\dagger. \quad (49)$$

Then for every $\mathbf{k} \neq 0$,

$$H_r(\mathbf{k}) \mathbf{w}(\mathbf{k}) = (a_r - 4b_r) \mathbf{w}(\mathbf{k}). \quad (50)$$

Thus this correction shifts the carrier without dispersing it.

Proof. Use $B^\dagger \mathbf{w} = 0$ and $S \mathbf{w} = -4 \mathbf{w}$. $\square$

In real space the same statement is the orthogonality of the two cellular Laplacians

$$L_2^\downarrow = \partial_2^\dagger \partial_2, \quad L_2^\uparrow = \partial_3 \partial_3^\dagger, \quad L_2^\downarrow L_2^\uparrow = L_2^\uparrow L_2^\downarrow = 0. \quad (51)$$

Every polynomial in $L_2^\downarrow$ and $L_2^\uparrow$ acts on the harmonic subspace by its constant term. A term proportional to $L_2^\uparrow$, however, disperses the cube-boundary component while leaving the three harmonic sheets pinned. Homology therefore distinguishes local boundaries from true harmonic cycles.

The proposition is not an unconditional all-orders theorem. The set of operators $BMB^\dagger$ is not a two-sided ideal in $\text{End}(C_2)$, and a general higher-order gauge-theory correction need not factor through links. Cubic symmetry alone permits carrier-projected fourth-order shapes built from

$$\begin{aligned} &1, \quad q, \quad e_2, \quad \frac{4e_2}{q}, \quad \frac{e_3}{q}, \\ &e_2 = \sum_{i < j} a_i a_j, \quad e_3 = a_1 a_2 a_3, \end{aligned} \quad (52)$$

where $a_i = 4 \sin^2(k_i/2)$. The rational shapes encode directional limits at Γ and need not belong to the boundary-factorized class.

7 Restricted cellular circuits

The static circuit language becomes rigorous only after its domain is specified. Let a reduced integral history define

$$x_h = \mathbf{e}_p + \sum_{j=1}^r \sigma_j \mathbf{e}_{f_j} - \tau \mathbf{e}_q \neq 0, \quad \partial_2 x_h = 0, \quad (53)$$

where the e_f are oriented unit basis vectors of $C_2(\mathbb{Z})$, $\sigma_j, \tau \in \{\pm 1\}$, and compatibility includes the retained endpoints, orientations, and sector constraints. Define

$$w_{\min} = \min \{ \|y\|_1 : 0 \neq y \in \ker_{\mathbb{Z}} \partial_2, y \text{ compatible} \}. \quad (54)$$

Proposition 9 (Weighted integral-circuit bound). *Every history satisfying Equation (53) obeys*

$$\boxed{r \geq w_{\min} - 2}. \quad (55)$$

Proof. By definition $w_{\min} \leq \|x_h\|_1$. The two endpoints contribute at most two units and the r insertions contribute at most r , so $\|x_h\|_1 \leq r + 2$. $\square$

The nonzero condition in Equation (53) is essential. Static-cancelling histories with $x_h = 0$, and histories whose balance closes only modulo N , fall outside the integral bound's domain. Every history with nonzero integral x_h remains subject to Proposition 9; exact Haar contraction, the Gram quotient, representation-dependent resolvents, sector projection, and retained-space folds are subsequent survival and amplitude filters, not exceptions to the bound. Hence an F -face unit-incidence cell boundary supplies the distinguished static candidate $r = F - 2$, not a universal lower bound or equality for physical transfer.

For a family of projected face-cell incidences $I_{r,a}$, define the zero-diagonal completion operator

$$(J_{r,a})_{fg} = - \sum_c (I_{r,a})_{fc} (I_{r,a})_{gc}^* \quad (f \neq g), \quad (56)$$

$$(J_{r,a})_{ff} = 0.$$

Then, entrywise,

$$J_{r,a} = D_{r,a} - I_{r,a} I_{r,a}^\dagger, \quad D_{r,a} = \text{diag}(I_{r,a} I_{r,a}^\dagger). \quad (57)$$

Only when $I_{r,a}$ is an actual cellular boundary restriction is its Gram operator an up-Hodge Laplacian.

Proposition 10 (Conditional completion escape). *Suppose all carrier-projected corrections below order r are momentum-independent scalars, and the complete non-cell and folded contribution at order r is also a momentum-independent scalar on the carrier. Suppose further that, modulo a momentum-independent carrier scalar, the full same-order cell contribution equals $K_r^{\text{cell}} = \sum_a c_{r,a} J_{r,a}$. If*

$$\text{spec}(P_{\text{car}}(\mathbf{k}) K_r^{\text{cell}}(\mathbf{k}) P_{\text{car}}(\mathbf{k}))$$

depends on $\mathbf{k}$, then $r_{\text{disp}} = r$. If only its scalar-free compression is nonzero, one may conclude first shape splitting but not necessarily dispersion.

Proof. Under the hypotheses, every lower-order term and every omitted same-order term changes only a momentum-independent scalar. The complete cell sum is therefore the first possible source of momentum-dependent carrier eigenvalues. Testing the sum, rather than an individual family, also permits cancellations between same-order cell operators. $\square$

8 Primitive simple-loop channel

There is a useful closed color factor inside the restricted history class for which every intermediate state remains a simple fundamental loop. Assume that each insertion has normalized matrix element $1/N$, that $E_{F,N} - E_j = -d_j C_F$ with nonzero d_j , that no proper retained-space return occurs, and that the signed ordering sum is exhaustive. We first use the formal $+V$ insertion convention. An ordering π then contributes

$$A_\pi = \frac{1}{N^r C_F^{r-1}} \prod_{j=1}^{r-1} \frac{-1}{d_j(\pi)}, \quad S_r = \sum_\pi \prod_{j=1}^{r-1} \frac{-1}{d_j(\pi)}. \quad (58)$$

Consequently this primitive subchannel factorizes as

$$\boxed{\begin{aligned} c_{r,\text{prim}}^{(+V)}(N) &= \frac{S_r}{N^r C_F^{r-1}} = \frac{2^{r-1} S_r}{N(N^2 - 1)^{r-1}}, \\ c_{r,\text{prim}}^{\text{phys}}(N) &= (-1)^r c_{r,\text{prim}}^{(+V)}(N). \end{aligned}} \quad (59)$$

For the large- N statement, keep r fixed and assume the ordering set and denominators $d_j(\pi)$ are rank-independent (or uniformly bounded away from zero with fixed cardinality), so $S_r = O(1)$. Then both conventions obey $c_{r,\text{prim}} = O(N^{-(2r-1)})$. The physical sign above follows from $H = H_E - uV$. The supplied cube ledger reports $S_4 = -20$, which specializes Equation (59) to the reported residue $c_N^\square = -160/[N(N^2 - 1)^3]$; the two sign conventions coincide at $r = 4$. This factorization concerns one distinguished simple-loop channel. It is not the full physical coefficient and does not establish a universal rank staircase or a first transfer order.

9 Reported axial datum

The axial channel can be isolated from the unresolved mixed-gradient invariant. Set

$$L(k) = 4 \sin^2 \frac{k}{2}. \quad (60)$$

On $\mathbf{k} = (k, 0, 0)$, the invariants in Equation (52) obey $q = L(k)$ and $e_2 = e_3 = 0$.

Reported computational result 11 (Cube and axial ledgers). *The supplied symbolic derivation ledger reports*

$$c_N^\square = -\frac{160}{N(N^2 - 1)^3}, \quad \boxed{\alpha_N = -4c_N^\square = \frac{640}{N(N^2 - 1)^3}}. \quad (61)$$

A supplied microscopic ledger further reports that, for $N = 3, \dots, 12$, cube localization and complete axial-channel assembly decompose the centered operator, modulo a momentum-independent scalar, as

$$K_{4,N}^{\text{axis}} = B M_{4,N} B^\dagger + c_N^\square H_\square, \quad (62)$$

where $H_{\square}$ is the axial cube-transfer operator. On the axial carrier vector, the same ledger reports the normalization

$$\mathbf{w}^{\dagger}(c_N^{\square}H_{\square})\mathbf{w} = \frac{\alpha_N}{4}L(k)^2. \quad (63)$$

These are exact-arithmetic formulas as reported. The symbolic history payload and the input/output artifacts for the microscopic sweep are not included in the present bundle; `verify_core.py` checks arithmetic consequences of the encoded formulas only.

Conditional corollary 12 (Axial fourth-order law). *Assume that the reported symbolic cube-history sum is correct and complete, and that the reported microscopic assembly exhausts the physical axial channel at a rank $N \in \{3, \dots, 12\}$. Then, for nonzero axial momentum,*

$$\lambda_{\text{car},N}^{(4)}(k, 0, 0) = c_{0,N} + \frac{\alpha_N}{4}L(k), \quad (64)$$

where $c_{0,N}$ is independent of k . In particular, $\alpha_3 = 5/12$ and the coefficient of $L(k)$ is $5/48$.

Proof. Because $B^{\dagger}\mathbf{w} = 0$, the first term in Equation (62) annihilates the carrier. By the reported normalization Equation (63), the raw cube-boundary numerator is $(\alpha_N/4)L(k)^2$, while $\|\mathbf{w}\|^2 = L(k)$. Division by the carrier norm gives the single power of $L(k)$ in Equation (64). $\square$

Under the assumptions of Conditional corollary 12, if those of Conditional corollary 7 also hold, the $SU(3)$ carrier therefore first disperses along an axial cut at order u^4 : for fixed nonzero k , the energy difference contains $(5/48)u^4L(k)$. The centered extension to Γ is continuous along the axis, although the carrier projector itself changes rank there.

Mixed invariants vanish on every axial cut. Consequently Conditional corollary 12 cannot determine their coefficient. Two supplied fourth-order records report the same axial datum but disagree on one off-axis mixed-gradient coordinate. Because their underlying artifacts are not both available here, their agreement is not counted as an independent reproduction. The target-blind linked contraction needed to adjudicate the discrepancy is incomplete. We therefore report neither a complete fourth-order carrier operator nor a fourth-order bandwidth, and no disputed off-axis value is used downstream.

10 Relation to prior flat-band constructions

It is useful to state precisely what is shared with, and what differs from, nearby constructions.

Line-graph and molecular-orbital models engineer a rectangular matrix D and obtain zero modes from $D^{\dagger}D$ or $DD^{\dagger}$ [5, 8, 9]. Here the same algebraic mechanism appears, but $D = \partial_2$ is selected by the orientation signs of charge-odd non-Abelian flux, and its prefactor is the exact representation-theoretic number t_N . The hopping

ratio is therefore derived from the gauge Hamiltonian rather than tuned as a lattice design.

The two-form spin construction of Chung and Gingras [10] also features plaquette variables, curl constraints, flat interaction bands, and winding sectors. Its microscopic degrees of freedom and emergent $U(1)$ description differ from the projected $SU(N)$ flux operator studied here. Hazra's contemporary face-graph construction [12] gives compact states and higher-cell generalizations using a designed face-incidence Hamiltonian. Our work was developed independently, and our new content relative to that framework is the non-Abelian strong-coupling derivation, the all-rank color coefficient, and the ledger-dependent $SU(3)$ third-order analysis.

The torus count likewise has a standard topological interpretation. Compact cube boundaries provide $L^3 - 1$ states, while singularity of the Bloch eigenvector forces noncontractible supplements [6, 7]. In the cubical gauge setting those supplements are exactly the three generators of $H_2(T^3)$.

11 Scope and physical interpretation

At Γ , the three face-orbital states transform as an axial vector of the cubic group. Together with parity even and charge conjugation odd, this gives the retained-space label T_1^{+-} . This label is an operator-space assignment. A physical glueball statement would require, at minimum, a volume-uniform dressed-operator overlap theorem, survival of the level after coupling to the complement of the one-plaquette external manifold, and a controlled continuum limit. None is implied by Theorem 4 and Conditional corollary 7.

There is nevertheless a clean operator-level bridge between the retained source and weak-field notation. Writing $U_{jk}(x) = e^{iX}$ with $\text{Tr } X = 0$,

$$B_i^-(x) = \text{Im Tr } U_{jk}(x) = -\frac{1}{6}\text{Tr } X^3 + \frac{1}{120}\text{Tr } X^5 + O(X^7). \quad (65)$$

This identifies two expansions of the same gauge-invariant source operator; it does not prove continuation of a pole, glueball identification, or a continuum limit.

Several limitations should therefore remain attached to every use of the result:

- The strong-coupling series is treated as a formal finite-order expansion; convergence is not asserted.
- The carrier is macroscopically degenerate at finite L , but its nearest incidence separation vanishes as L^{-2} .
- The all-rank unconditional factorization stops at second order. Third-order $SU(3)$ rigidity is conditional on the supplied linked ledger and lifter census. The reported cube formula is all-rank, while the physical axial conclusion is conditional on the supplied

$N = 3, \dots, 12$ microscopic sweep in Conditional corollary 12.

- The ordinary vertex Gauss law is already satisfied by the Wilson-loop basis. The additional equation $\partial_2 c = 0$ is a boundary-free two-chain condition.
- No disputed off-axis fourth-order coefficient, historical fifth-order series, continuum extrapolation, or mass-ratio prediction enters the theorem.

Within those boundaries, the self-contained second-order result supplies a controlled operator seed whose geometry, degeneracy, and all-rank color scaling are known. The higher-order ledger values and their conditional consequences are recorded separately.

12 Reproducibility and evidence boundary

The incidence, homology, fixed-order locality, and all-rank second-order arguments are self-contained. The $SU(3)$ third-order totals, the 32-class lifter census, the symbolic cube formula, and the $N = 3, \dots, 12$ physical axial sweep are reported by supplied exact-arithmetic derivation ledgers. Their underlying enumeration and certificate artifacts are not included in this bundle.

A portable standard-library Python program distributed with the source, `verify_core.py`, checks identities among encoded rational coefficients, the incidence identity at deterministic generic momenta, the chain condition $\partial_2 \partial_3 = 0$, and finite-torus ranks for $L = 1, \dots, 4$. It does not derive the third-order linked coefficients, enumerate the 32 lifter classes, sum the cube histories, or reproduce the microscopic axial sweep. Its third- and fourth-order checks are arithmetic regressions only.

The supplied repository ledger identifies commit `ca3d440a7f93c17569e12d0511847505b6b72c5a` and reports 119 exact-rational checks, 29 numerical cross-checks, 349 repository tests, and 28 Lean theorems with no omitted proofs. The corresponding logs and omitted certificate payloads do not accompany this bundle, so the counts are provenance information rather than independent evidence here. Only a subset bears directly on the present document. The relevant commands are

```
uv sync --frozen --all-extras
make verify
make check
make lean
```

and the paper-level checks can be reproduced with

```
python3 verify_core.py
```

Machine checks certify algebra relative to stated definitions and inputs; they do not replace the perturbative derivation, and they do not certify an uncompleted off-axis fourth-order linked contraction. The additional cold

$SU(3)$ resolvent engine used for Equations (40) and (41) produces those exact rational kernels but imports the complete third-order coefficient as a regression target; it is not an independent cold computation of the direct third-order chain. A separate pentagonal-prism program supplied with the audit deliberately sets every unbalanced (p, q) , $p \neq q$, Haar sector to zero, including the physical $SU(3)$ determinant sectors. It is a negative control only, and none of its hopping coefficients enters this paper.

13 Conclusions

Strong-coupling $SU(N)$ dynamics and cubical topology separate self-containedly through second order in the charge-odd one-plaquette sector. The non-Abelian shared-link calculation supplies the positive all-rank coefficient t_N , while the oriented boundary map supplies the Gram operator $BB^\dagger$. Its kernel is a homological flat carrier of boundary-free two-chains, with cube-boundary compact states and three wrapping-sheet generators on the torus.

The master framework explains how to continue: select the complete electric eigenspace, retain temporal order, quotient Gram-null networks, apply representation-resolved resolvents, assemble Feshbach folds, perform rooted linked subtraction, and only then distinguish scalar position, internal shape, and momentum dispersion. At every fixed order this reduces to finite algebra under the stated locality and finite-reach assumptions. The restricted circuit and primitive-channel formulas identify useful cellular subchannels but are not universal transfer laws.

Beyond second order, this document records rather than independently reconstructs the controlling finite computations. Supplied ledgers report the complete $SU(3)$ third-order entries and a zero census for 32 candidate lifter classes. Conditional on that ledger, the incidence algebra survives through order u^3 . Supplied ledgers also report $\alpha_N = 640/[N(N^2 - 1)^3]$ and its complete axial identification for $N = 3, \dots, 12$. Conditional on that microscopic closure, $\alpha_3 = 5/12$ yields axial lifting at order u^4 .

Independent regeneration of the omitted third- and fourth-order payloads is a reproducibility task; the off-axis mixed-gradient contraction is a separate unresolved physics calculation. No complete fourth-order bandwidth, continuum extrapolation, glueball identification, or mass-gap claim follows here. The unconditional result is the dynamically generated all-rank homological flat band through second order, with all higher-order claims explicitly graded.

Table 1: Master claim and evidence map. The phrase *exact arithmetic* means that a displayed value is represented rationally or symbolically; it does not by itself establish that the underlying history universe was complete.

Claim	Scope	Evidence/check
$BB^\dagger = qI - \mathbf{w}\mathbf{w}^\dagger$ and $\text{spec}(BB^\dagger) = \{0, q, q\}$ $\dim \ker \partial_2 = L^3 + 2$	all $\mathbf{k}$	Self-contained. Direct multiplication; deterministic generic-momentum residuals in <code>verify_core.py</code> Self-contained. Homology proof; exact modular ranks for $L = 1, \dots, 4$
Fixed-order support reduction	periodic cubical complex local, finite root; fixed r	Self-contained under stated hypotheses. Proposition 1; finite Haar–Gram quotient and explicit fold metadata required
Weighted integral-circuit bound	nonzero integral unit-incidence histories	Self-contained within its restricted domain. Proposition 9; not a physical mobility law
Primitive simple-loop factorization	fixed r ; rank-controlled simple-loop histories	Conditional/restricted. Equation (59); requires exhaustive orderings and the displayed intermediate-state assumptions
$t_N = \mathcal{B}_N - \mathcal{A}_N > 0$	integer $N \geq 3$	Self-contained. Channel-projector derivation and rational simplification
Cold $PWRWP$ and PWR^2WP kernels	local $SU(3)$ scope	Reproduced partial calculation. Cold-derived electric spectra and rational kernels; the full u^3 coefficient is only a regression target and $PWRWP$ is not cold-computed
Complete $SU(3)$ scalar and hopping entries through u^3	projected charge-odd sector	Ledger-reported. Exact rationals transcribed from the supplied linked ledger; the verifier checks their assembly but does not regenerate the contractions
Third-order 32-class lifter census	$SU(3)$	Ledger-reported. The supplied record reports 32 exhaustive determinant-aware classes with zero contraction; the class payload and per-class outputs are not included
Third-order incidence factorization	$SU(3)$	Conditional. Follows from the self-contained incidence theorem if the reported census is exhaustive and the linked coefficients are complete
$c_N^\square = -160/[N(N^2 - 1)^3]$ and $\alpha_N = -4c_N^\square$	reported all-rank cube formula	Ledger-reported. The bundled verifier checks arithmetic consequences of the encoded expression, not the symbolic history sum
Identification of α_N with the complete axial coefficient	reported sweep $N = 3, \dots, 12$	Conditional. A supplied microscopic ledger reports the complete sweep, but its enumeration and certificate artifacts are not included
Complete order-four carrier operator	off-axis $SU(3)$	Open. Two supplied branches disagree on one mixed-gradient coordinate; no band or bandwidth reported

A Channel weights in closed form

For completeness, substituting Equation (27) into Equation (26) gives

$$\begin{aligned}
w_1 &= -\frac{2}{N(N-1)(N+1)}, \\
w_{\text{Adj}} &= -\frac{2(N-1)(N+1)}{N(2N^2-1)}, \\
w_{\Lambda^2 F} &= -\frac{N-1}{(N+1)(2N-3)}, \\
w_{\text{Sym}^2 F} &= -\frac{N+1}{(N-1)(2N+3)}. \tag{66}
\end{aligned}$$

The mixed and like sums are

$$\mathcal{A}_N = w_1 + w_{\text{Adj}}, \quad \mathcal{B}_N = w_{\Lambda^2 F} + w_{\text{Sym}^2 F},$$

which simplify to Equations (28) and (29). Their difference has numerator $2N(N^2 - 4)$, proving both the $SU(2)$ zero and positivity for $N \geq 3$. Expanding in $z = 1/N^2$ gives

$$t_N = \frac{1}{4N^3} + \frac{1}{N^3} \left(-\frac{z}{16} - \frac{77z^2}{64} - \frac{1021z^3}{256} + \dots \right).$$

B Finite-volume chain calculation

The cubical boundary of an oriented (i, j) face based at x , with $i < j$, is

$$\partial_2[x; i, j] = [x + \mathbf{e}_i; j] - [x; j] - [x + \mathbf{e}_j; i] + [x; i]. \tag{67}$$

The boundary of a cube is

$$\begin{aligned}\partial_3[x; 1, 2, 3] &= [x + \mathbf{e}_1; 2, 3] - [x; 2, 3] \\ &\quad - [x + \mathbf{e}_2; 1, 3] + [x; 1, 3] \\ &\quad + [x + \mathbf{e}_3; 1, 2] - [x; 1, 2].\end{aligned}\quad (68)$$

Substitution of Equation (68) into Equation (67) cancels every edge pairwise. On the torus, $\ker \partial_3$ is generated by the sum of all oriented cubes, so $\text{rank } \partial_3 = L^3 - 1$. Since $H_2(T^3) \simeq \mathbb{R}^3$, the count in Theorem 3 follows.

C Reported $SU(3)$ coefficient ledger

The coefficient assembly used in Conditional corollary 7 is summarized in Table 2. All entries are rational.

Table 2: $SU(3)$ charge-odd coefficient ledger.

Order	Quantity	Value
u^0	electric plaquette	$8/3$
u^1	scalar	1
u^2	$BB^\dagger$ coefficient	$5/612$
u^2	flat scalar	$11/306$
u^3	$BB^\dagger$ coefficient	$1975/124848$
u^3	per-neighbor leakage	$-12331/249696$
u^3	flat scalar	$-109151/249696$

The last line is the exact identity

$$-\frac{109151}{249696} = \frac{7}{32} + 12 \left(-\frac{12331}{249696} \right) - 4 \left(\frac{1975}{124848} \right).$$

D Data and source availability

The manuscript source and the portable core verifier accompany this version. The source bundle also contains a disposition note for the additional derivation and backend files used in this revision. The raw 32-class contraction payload, symbolic cube-history payload, microscopic rank-sweep artifacts, and cited continuous-integration logs do not accompany this document. The private repository commit in Section 12 is therefore a provenance locator, not reader-accessible reproducible evidence by itself. The unresolved off-axis fourth-order contraction is explicitly excluded from the results reported here.

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